

Preparation, thermal characterization and examination of phase change materials (PCMs) enhanced by carbon-based nanoparticles for solar thermal energy storage



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ABSTRACT

This paper presents the preparation and thermal characterization of phase change materials (PCMs) enhanced by carbon-based nanoparticles, including graphene nanoplatelets (GNPs), multi-walled carbon nanotubes (MWCNTs) and nano-graphite (NG). A systematic experimental framework, consisting of material selection and preparation, material property characterization and thermal performance examination, was proposed and used in this study to facilitate the development of nano-enhanced PCMs (NePCMs) for solar thermal energy storage applications. By applying this framework, the characteristics and potential performance of PCM composites can be comprehensively understood, and better assessed before practical applications. It was found that the thermal conductivity of the myristic acid (MA) can be significantly enhanced by adding the nanoparticles in particular GNPs as additives into the PCM. The thermal conductivity of the PCM composites can be improved by 176.26%, 47.30% and 44.01% respectively under the solid phase, by adding GNPs, MWCNTs and NG with a concentration of 3 wt%. However, the concentration of the nanoparticles needs to be carefully determined to maximise the benefit in thermal conductivity enhancement. Different from that under the solid phase, the thermal conductivity enhancement of the NePCMs developed under the liquid phase followed linear increasing trends with relatively low increasing rates, when increasing the concentration of the nanoparticles. Besides the thermal conductivity enhancement, the adding of nanoparticles also modified the phase change process with a smaller phase change temperature range and eliminated supercooling while maintaining the high latent heat capacity. A further thermal performance examination demonstrated that the prepared NePCMs showed high thermal and chemical stability, which can be used to substantially reduce the phase transition time, and therefore are good potential candidates for solar thermal energy storage applications.

1. Introduction

Due to increasingly excessive energy consumption and severe environmental problems over the last two decades, the development and deployment of thermal energy storage (TES) systems using phase change materials (PCMs) to facilitate energy savings, renewable energy utilization, and efficiency enhancement of thermal systems have attracted considerable attention [1–3]. PCMs are substances that can absorb, store and release a large amount of thermal energy at relatively constant temperatures [4,5]. One of typical applications of PCM TES is for the storage of solar thermal energy which is considered to be the most abundant renewable energy [6]. The utilization of PCM TES can overcome the intermittency of solar energy, thereby providing a better

solution to rationalise the utilisation of solar thermal energy, compared with sensible heat TES systems [7].

For proper functioning of a PCM TES system, rapid melting and solidification of the PCMs are desirable to respond to fast load variations. However, most PCMs have the inherent disadvantage of low thermal conductivity, which results in low charging and discharging rates, and limits the overall energy storage capacity and efficiency of PCM TES systems [8,9]. To solve this problem, a number of heat transfer enhancement techniques have been proposed, such as the use of fins and heat pipes, embedment/dispersion or shape-stabilisation using high thermal conductivity materials, micro-encapsulation and macro-encapsulation [10–14]. The dispersion of high thermal conductivity additives is one of the effective approaches to enhancing the

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thermal conductivity of PCMs, however, normal millimetre-sized or micron-sized additives have the disadvantages of relatively low surface energy and relatively large density, which are more likely to cause sediments [15]. As an alternative solution, adding nanoparticles into PCMs to prepare PCM composites with high thermal conductivity is currently of great interest and receiving increasing attention [16–18]. For instance, Harikrishnan and Kalaiselvam [19] prepared a nano-enhanced PCM composite using CuO nanoparticles as additives and oleic acid as the base PCM. It was found that by adding 2 wt% of CuO nanoparticles, the thermal conductivity of the PCM composite can be improved significantly by 98.66%. As a result, the time needed for solidification and melting processes can be reduced by 27.67% and 28.57%, respectively. Zeng et al. [20] added copper nanowires into tetradecyl alcohol to prepare nano-enhanced PCM composites. It was found that by adding 11.9 vol% copper nanowires, the thermal conductivity reached as high as $2.86 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, which was about 9 times that of the pure tetradecyl alcohol. Sahan and Paksoy [21] used synthesized Fe_3O_4 nanoparticles as additives to prepare nano- Fe_3O_4 /paraffin PCM composites by a sol-gel method. It was interesting to find that the adding of nano- Fe_3O_4 can increase both thermal conductivity and thermal storage capacity. The Differential Scanning Calorimetry (DSC) test results showed that the latent heat storage capacity increased by approximately 20% with the adding of 10 wt% nano- Fe_3O_4 . Zeng et al. [22] prepared nano-enhanced PCM composites by adding Ag nanowires into 1-Tetradecanol. The thermal characterization showed that the thermal conductivity of the PCM composite was more than 3.5 times higher than that of the raw PCM, when adding 11.8 vol% of Ag nanowires.

Besides metallic and metallic oxide nanoparticles, carbon-based nanoparticles with extremely high thermal conductivity (e.g. $3000\text{--}6600 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$) [23–25] have been frequently used in many studies [26–35] as additives to enhance the thermal conductivity of PCMs, especially for organic PCMs which generally have lower thermal conductivity compared with inorganic PCMs. For instance, Ji et al. [26] added functionalized multi-walled carbon nanotubes (MWCNTs) into palmitic acid to modify its thermal properties. It was found that the well-dispersed MWCNTs in palmitic acid can effectively improve the thermal conductivity of the nano-enhanced PCM composites. Sahan et al. [27] used activated carbon (AC) and MWCNTs as additives to prepare AC/paraffin and MWCNTs/paraffin composites with better thermal properties, respectively. The thermal characterization showed that the thermal conductivity of the MWCNTs/paraffin and the AC/paraffin composites increased by 38.5% and 17.2% respectively, by adding 1 wt% and 10 wt% of the nanoparticles. The increase in the latent heat (by 9.6%) was also observed when adding 1 wt% MWCNTs into the paraffin, compared with that of the raw paraffin. Choi et al. [28] studied and compared the enhancement in thermal conductivity when adding three different carbon nanoparticles to stearic acid. The comparison showed that with adding 0.1 vol% MWCNTs, graphite and graphene, the corresponding thermal conductivities of the mixtures increased by 10.5%, 9.91% and 21.5%, respectively. Li [29] improved the thermal conductivity of paraffin by adding nano-graphite (NG). Random dispersion of the NG was observed in the paraffin-NG composites, and its thermal conductivity reached $0.9362 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ when the NG concentration was 10 wt%, which was 7.41 times that of the raw paraffin. Yuan et al. [30] used graphene nanoplatelets (GNPs) and expanded graphite (EG) as additives to modify the thermal conductivity of palmitic-stearic acid (PA-SA). It was found that although both GNPs and EG can effectively improve the thermal conductivity of PA-SA linearly, by adding 8 wt% of EG, the thermal conductivity was 15.8 folds higher than that of the raw PA-SA, and 2.7 folds higher than that using 8 wt% GNPs as additives. Seki et al. [31] modified the thermal properties of adipic acid (AA) and sebacic acid (SA) eutectic by adding GNPs as additives. With different loading rates of 0.1 wt%, 0.3 wt% and 0.5 wt%, the thermal conductivity of AA-SA can be improved by 6%, 11% and 19% and the supercooling degree decreased from 17.7°C to 6.8°C ,

3.9°C and 3.6°C , respectively. Colla et al. [32] selected carbon black (CB) nanoparticles and paraffin to prepare nano-enhanced PCMs. It was found that CB based nano-PCMs showed high thermal conductivity enhancement of more than 25%. Wang et al. [33] dispersed treated carbon nanotubes (CNTs) into the palmitic acid (PA) for property modification. The thermal characterization showed that the thermal conductivity of prepared treated CNTs/PA composite was 46.0% higher than that of the raw PA, when the concentration of treated CNTs was 1 wt%.

This paper presents the preparation and thermal characterization of high-performance nano-enhanced PCM composites for solar thermal applications using myristic acid (MA) and carbon-based nanoparticles, including GNPs, MWCNTs and NG. The main research idea of this work is to develop PCM composites with desired characteristics, especially with high thermal conductivity, to facilitate the effective thermal energy management of solar thermal systems. The novelty of this study is the development and utilisation of a systematic experimental framework to guide material preparation, properties characterisation and performance examination of nano-enhanced PCMs (NePCMs). Accordingly, the base PCM and different nanoparticles were selected to develop various NePCM composites. The chemical and thermal properties of the prepared PCM composites were characterized by scanning electronic microscope (SEM) analysis, Fourier transform infrared (FT-IR) spectroscopy, thermal conductivity measurement, DSC test and Thermogravimetry (TG) analysis. The thermal performance of the NePCM composites was further examined by cooling bath and thermal cycling stability analysis.

2. Research methodology

The overall research methodology used in this study was refined based on a general framework proposed in a previous study [36], as presented in Fig. 1. It mainly consists of three steps, including material selection and preparation (Step 1), material property characterization (Step 2) and thermal performance examination (Step 3).

2.1. Materials selection and preparation of nano-enhanced PCM composites

2.1.1. Materials selection

To facilitate solar thermal applications (e.g. solar hot water, and solar powered desiccant cooling etc.) of the nano-enhanced PCMs, a commercial myristic acid (MA) with a nominal phase change temperature of $54\text{--}55^\circ\text{C}$ supplied from Aladdin biochemical technology Co. Ltd. was selected and used as based PCM. MA was selected as the base PCM mainly due to its suitable phase change temperature for low-grade solar thermal storage applications. In addition, as one of the typical fatty acids, it also has the advantages of almost no supercooling and long-term stability without phase segregation. Three different types of carbon-based nanometre-sized materials, including GNPs, MWCNTs and NG were used as high thermal conductive additives, respectively. The properties of the base PCM and the nano-additives are summarized in Table 1. It is worthwhile to mention that all chemicals used were received from the suppliers without further functionalisation.

2.1.2. Preparation of NePCM composites

The two-step method which has been extensively used in the preparation of nanofluids [37,38] was adopted to prepare the nano-enhanced PCM composites in this study. Firstly, MA was completely melted above 70°C for 10 min. The three types of carbon-based nano-additives were then added and mixed with the molten MA with different weight concentrations individually following the below procedure: magnetic stirring for 30 min first, and followed by the ultrasonic vibration for 1 h to facilitate a homogeneous dispersion; lastly, the nano-enhanced PCM samples were cooled to the ambient temperature. The corresponding schematic diagram is shown in Fig. 2. The experimental plan designed to prepare the nano-enhanced PCM composite

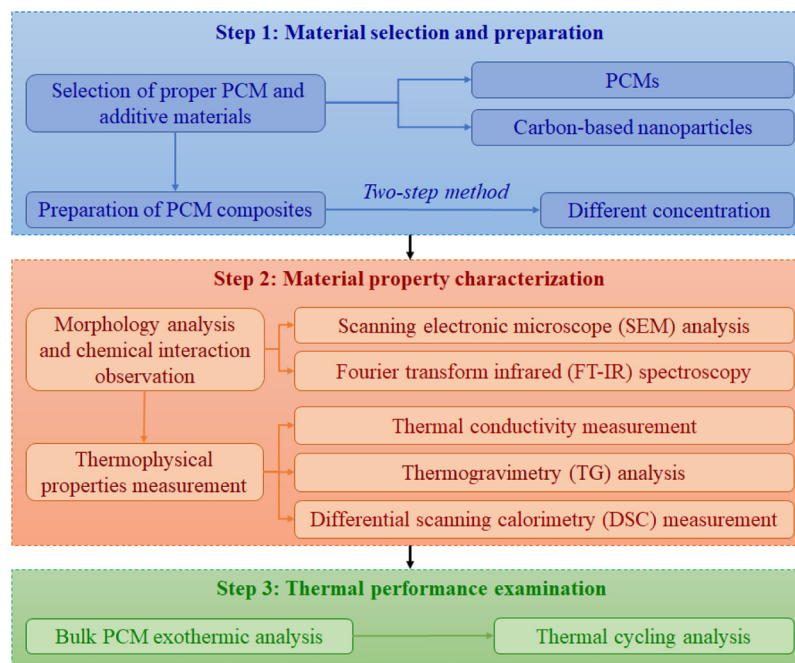


Fig. 1. The NePCMs development framework used in this study.

samples is summarized in Table 2.

2.2. Material property characterization

The characterization of the PCM composites involved the morphology analysis by SEM, the chemical interaction examination by FT-IR spectroscopy, and the measurements of the thermal physical properties including thermal conductivity, enthalpy-temperature relationship, and thermogravimetry feature.

2.2.1. SEM analysis and FT-IR spectroscopy

The microstructures of the carbon-based nano-enhanced materials, including 3 wt% GNPs/MA, 3 wt% MWCNTs/MA and 3 wt% NG/MA PCM composites were observed using a S-4800 SEM (from Hitachi, Japan). The interactions between the carbon-based nano-additives and the based PCM were characterized by using Excalibur 3100 FT-IR spectrometer (from Varian, USA), with KBr pellets. The FT-IR spectrums were recorded in the wavenumber range of 4000–400 cm^{-1} .

2.2.2. Measurement of thermal physical properties

The thermal conductivities of the prepared PCM composites, and the raw MA, were tested based on the transient hot-wire method by a thermal conductivity meter TC 3000E (from Xi'an XIATECH, China). The measurement accuracy of the thermal conductivity meter TC 3000E used was $\pm 3\%$. The thermal conductivity of these samples was measured under both solid and liquid phases. For the solid phase, the samples were first melted and filled into moulds with a size of 5 cm

(length) $\times$ 3.5 cm (width) $\times$ 0.2 cm (height). After cooled to the room temperature and solidified, the cuboid-shaped samples were polished in the upper and lower surfaces to provide two smooth surfaces for the installation of the sensor. For the liquid phase, each sample was melted and poured into a 50 ml beaker and kept in a water bath at 70 $^{\circ}\text{C}$, while the sensor was inserted into the centre of the beaker to test the thermal conductivity under liquid phase. The measurement of each sample under solid or liquid phases was repeated for three times to obtain the average values. For MA, 3 wt% GNPs/MA, 3 wt% MWCNTs/MA and 3 wt% NG/MA, their thermal conductivity was further measured under the sample temperatures of 35 $^{\circ}\text{C}$, 45 $^{\circ}\text{C}$ and 60 $^{\circ}\text{C}$.

The enthalpy-temperature relationship of the prepared samples was measured by using a DSC (DSC1 from Mettler Toledo, Switzerland) with a temperature scanning rate of 10 $^{\circ}\text{C min}^{-1}$ under a constant nitrogen atmosphere of 50 ml min^{-1} . This DSC equipment used a so-called FRS 5 + DSC sensor, whose TAWN resolution and sensitivity were reported as 0.12 and 11.9, respectively. It was calibrated using standard Indium and Zinc samples provided by the supplier. In the DSC tests, aluminium pans were used as the sample holders and the mass of each sample was controlled within 5 mg. The resulted DSC curves within the temperature range of 0–100 $^{\circ}\text{C}$ were analysed in detail.

TG curves of the prepared samples were recorded by a thermogravimeter (TGA1 from Mettler Toledo, Switzerland) under nitrogen stream at a flow rate of 30 ml min^{-1} . The measurement accuracy of this thermogravimeter was $\pm 0.5^{\circ}\text{C}$. The rate of heating was 10 $^{\circ}\text{C min}^{-1}$ in the temperature range of 30–500 $^{\circ}\text{C}$.

Table 1
Main properties of MA and additives.

Substances	Purity	Density or tapped density	Main physical parameters	Manufacturers
MA	98%	0.898 g cm^{-3}	Melting point: 54–55 $^{\circ}\text{C}$ Latent heat of fusion: 194.90 J g^{-1}	Aladdin biochemical technology Co. Ltd.
GNPs	99.5%	0.075 g cm^{-3}	Thickness: 3–10 nm Diameter: 5–10 μm	Nanjing XFNANO Materials Tech Co. Ltd
MWCNTs	> 95%	0.270 g cm^{-3}	Length: 10–30 μm Diameter: < 8 nm	Nanjing XFNANO Materials Tech Co. Ltd
NG	99.9%	0.252 g cm^{-3}	Thickness: < 40 nm Diameter: 3–6 μm	Nanjing XFNANO Materials Tech Co. Ltd

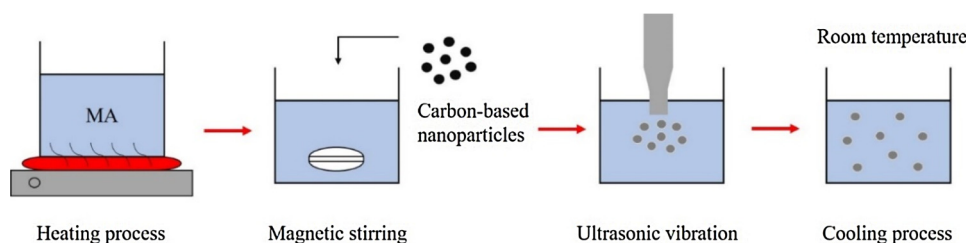


Fig. 2. Schematic diagram of the sample preparation process.

Table 2

Prepared NePCM composites with different weight concentrations of the additives.

Sample index	Sample	Additive concentration
0	MA	–
1	GNPs/MA	1 wt%
2	GNPs/MA	2 wt%
3	GNPs/MA	3 wt%
4	MWCNTs/MA	1 wt%
5	MWCNTs/MA	2 wt%
6	MWCNTs/MA	3 wt%
7	NG/MA	1 wt%
8	NG/MA	2 wt%
9	NG/MA	3 wt%

2.3. Thermal performance examination

2.3.1. Exothermic performance analysis for bulk samples

It is of great interest to investigate the thermal performance of bulk PCMs or PCM composites, due to the fact that it may be different from that tested based on small samples. The exothermic performance of the bulk NePCM composites or raw MA was tested in this study using a water bath test apparatus, CC-K6 with Pilot ONE (from Huber, German). It consists of a water jacketed glass vessel, a PT100 temperature sensor and a water bath thermostat, as shown in Fig. 3. The melted MA or NePCM composites were filled in the water jacketed glass vessel, while the glass vessel was supplied with the water flow of 30 °C from the water bath thermostat. The accuracy of the PT100 temperature probe used is ± 0.02 °C. It was placed in the centre of the vessel, and the temperature measurement interval was 5 s.

2.3.2. Thermal cycling test

The thermal cycling stability, which is an important indicator to assess the practical applicability of a PCM or PCM composite, was tested through accelerated thermal cycling test in this study. In the test, the prepared NePCM composite samples were encapsulated into 2 ml centrifuge tubes, which were melted in the hot water bath at 70 °C and then solidified at 30 °C periodically. The phase change characteristics of the PCM composites were measured by the DSC after per 100 thermal cycles, while their crystalline and chemical structures before and after 300 thermal cycles were characterized using an X-ray diffraction (XRD) (D8 focus from Bruker, Germany), in a diffraction angle (2θ) ranging from 5°–90° and FT-IR in the wavenumber range of 4000–400 cm^{-1} ,

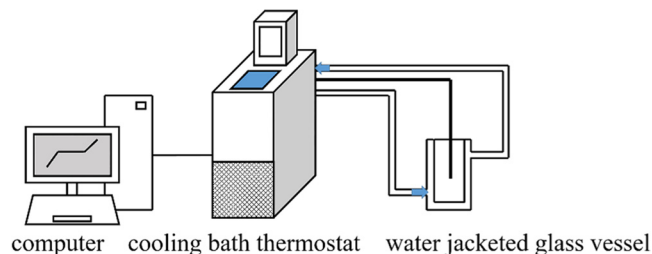


Fig. 3. Apparatus of exothermic experiments.

respectively.

3. Results and discussion

3.1. Microscopic structure

The microstructures of the PCM composites and carbon-based nano-additives are presented in Fig. 4. It can be seen from Fig. 4a that GNPs characterized a glossy plane structure with the high surface-to-thickness ratio. The GNPs seemed to be agglomerate, however, after mixed with MA through magnetic stirring and ultrasonic vibration, the GNPs seemed to be dispersed in the PCM, as can be seen in Fig. 4b. The plate-like structure of GNPs was maintained in the GNPs/MA composite, which facilitated the formation of a functional GNPs thermal conduction network. The MA was introduced into the network and covered GNPs, making them look thicker (see Fig. 4b), similar to that reported by Li [29]. Compared with that of the GNPs, the high aspect ratio of MWCNTs was achieved due to their long tubular structure, as can be seen in Fig. 4c. This long tubular structure with a high aspect ratio also enabled the MWCNTs to interlink with each other and resulted in a thermal conductive network after mixed with MA as shown in Fig. 4d. As MA penetrated and filled the network of MWCNTs quite well, the composite looked homogeneous, indicating an acceptable dispersion of the MWCNTs, but some tips of MWCNTs still can be observed clearly. Different from GNPs and MWCNTs, the aspect ratio of NG was relatively lower (see Fig. 4e). As a consequence, the NG/MA composite showed a weak connection between the NG, as can be seen in Fig. 4f. The agglomeration of NG can be clearly seen from Fig. 4e, but after prepared as NG/MA composite, a good dispersion of NG seemed to be achieved, as presented in Fig. 4f. It is worthwhile to note that even though acceptable dispersion of the nanoparticles in the PCM seemed to be achieved through a rational composite preparation process (e.g. using ultrasonic vibration), agglomerations of nanoparticles can still be found in the PCM composites, especially for GNPs/MA and MWCNTs/MA (see Appendix A for details).

3.2. FT-IR analysis

Fig. 5a presents the FT-IR spectrums of MA, GNPs and GNPs/MA composites. A series of characteristic absorption peaks can be observed in the FT-IR spectrum of MA. For instance, the absorption peaks at around 2920 cm^{-1} and 2850 cm^{-1} were caused by the stretching vibrations of the groups $-\text{CH}_3$ and $-\text{CH}_2$ in MA, while the characteristic absorption peak at around 1700 cm^{-1} represented the group $\text{C}=\text{O}$ stretching vibration. The peak of the bending vibration of $-\text{CH}_2$, located at around 1460 cm^{-1} , and that of the C–H and C–C chemical bonds laid at around 1300 cm^{-1} . The absorption peaks of rocking vibration and bending vibration can also be observed at around 933 cm^{-1} and 721 cm^{-1} , respectively. Differently, no obvious absorption peak in the FT-IR spectrum of GNPs was observed. It can also be seen from Fig. 5a that the FT-IR spectrums of the GNPs/MA composites were consistent with that of the raw MA, which contained the same characteristic absorption peaks in MA without the emergence of any apparently new peak. It meant that no considerable chemical interaction

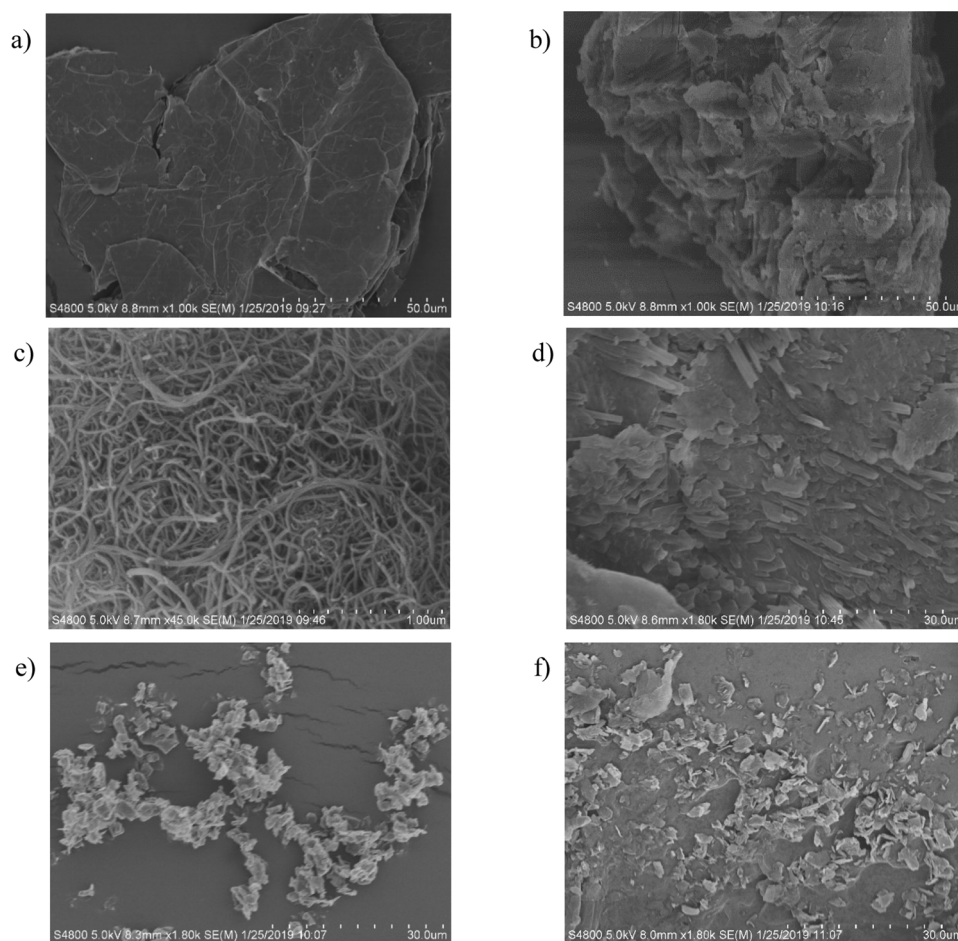


Fig. 4. Microscopic structures for: a) GNPs; b) 3 wt% GNPs/MA; c) MWCNTs; d) 3 wt% MWCNTs/MA; and e) NG; f) 3 wt% NG/MA.

occurred between MA and GNPs, indicating that the GNPs/MA composites prepared were chemically stable. Similar to that of GNPs and MA in the GNPs/MA composites, no considerable chemical interaction was observed between the nano-additives and MA according to the FT-IR spectrums of MWCNTs and MWCNTs/MA, and NG and NG/MA shown in Fig. 5b and c, respectively.

3.3. Thermal conductivity analysis

The variation of the thermal conductivity of the nano-enhanced PCM composites with the change of the concentrations of different carbon-based nano-additives is presented in Fig. 6, and the percentages of enhancement in thermal conductivity are further summarized in Table 3. The standard error in thermal conductivity measurements was only up to $0.0024 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$. It can be seen that the thermal conductivity of the GNPs/MA composites in both solid and liquid phases increased when increasing the weight concentration of GNPs (Fig. 6a). In the solid phase, the thermal conductivity of 1 wt% GNPs/MA composite reached $0.4405 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$, which was improved by 101.51% in comparison with that of the raw MA. By further increasing the GNPs concentration to 2 wt% and 3 wt%, the relative increments in thermal conductivity reached by 127.04% and 176.26%, respectively. However, the increasing rate tended to reduce with the increase of the GNPs concentration. It can be attributed to the agglomeration or even sedimentation phenomena which were likely to occur in the PCM composites under high nanoparticle concentrations. As a consequence, the total amount of functional nanoparticles which were beneficial to the thermal conduction enhancement cannot be improved proportionally, when further increasing the weight concentration of the nanoparticles.

A similar phenomenon was reported and analysed by Harikrishnan and Kalaiselvam [19], Wu et al. [39] and Choi et al. [28]. Different from the nonlinear increasing trend in the solid phase, the thermal conductivity of the GNPs/MA composite under the liquid phase experienced a linear increasing trend with the increasing of the GNPs concentration. However, its corresponding values were much lower than that of the solid phase. For instance, 3 wt% GNPs only contributed to the improvement in the thermal conductivity of 59.70% for the GNPs/MA composite under the liquid phase, compared to that of 176.26% under the solid phase.

Similar to the GNPs/MA composites, the thermal conductivity of the MWCNTs/MA composites increased with the increase of MWCNTs concentration, following a nonlinear trend under the solid phase while a linear trend under the liquid phase, as can be seen in Fig. 6b. When adding 1 wt% MWCNTs into MA, the thermal conductivity of the MWCNTs/MA composite can be improved substantially by 33.71% in the solid phase and 3.63% in the liquid phase, respectively. By further increasing the MWCNTs concentration to 3 wt%, the thermal conductivity reached $0.3220 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$ (for solid phase) and $0.2083 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$ (for liquid phase), which were 47.30% and 12.84% higher than that of the raw MA, respectively.

Even though similar trends were found, the thermal conductivity of the MWCNTs/MA composite was obviously lower than that of the GNPs/MA composite. It can be explained by the heat transfer mechanism within nano-enhanced PCMs. The enhancement in thermal conductivity resulted from the high thermal conductive percolation network established by the nanoparticles in the NePCM composites, which enabled the efficient transport of phonons [40]. However, this enhancement was limited by the heat transfer across the interfaces

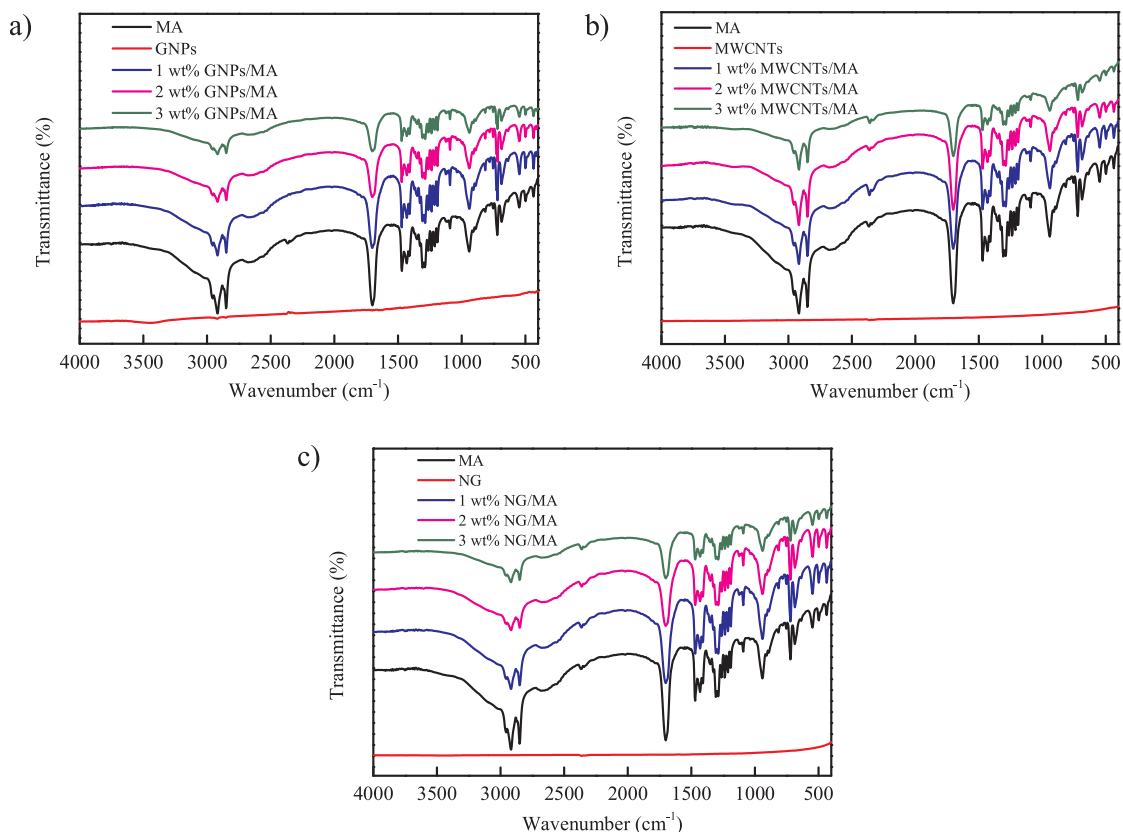


Fig. 5. FT-IR spectra for: a) MA, GNPs and GNPs/MA PCM composites; b) MA, MWCNTs and MWCNTs/MA PCM composites; and c) MA, NG and NG/MA PCM composites.

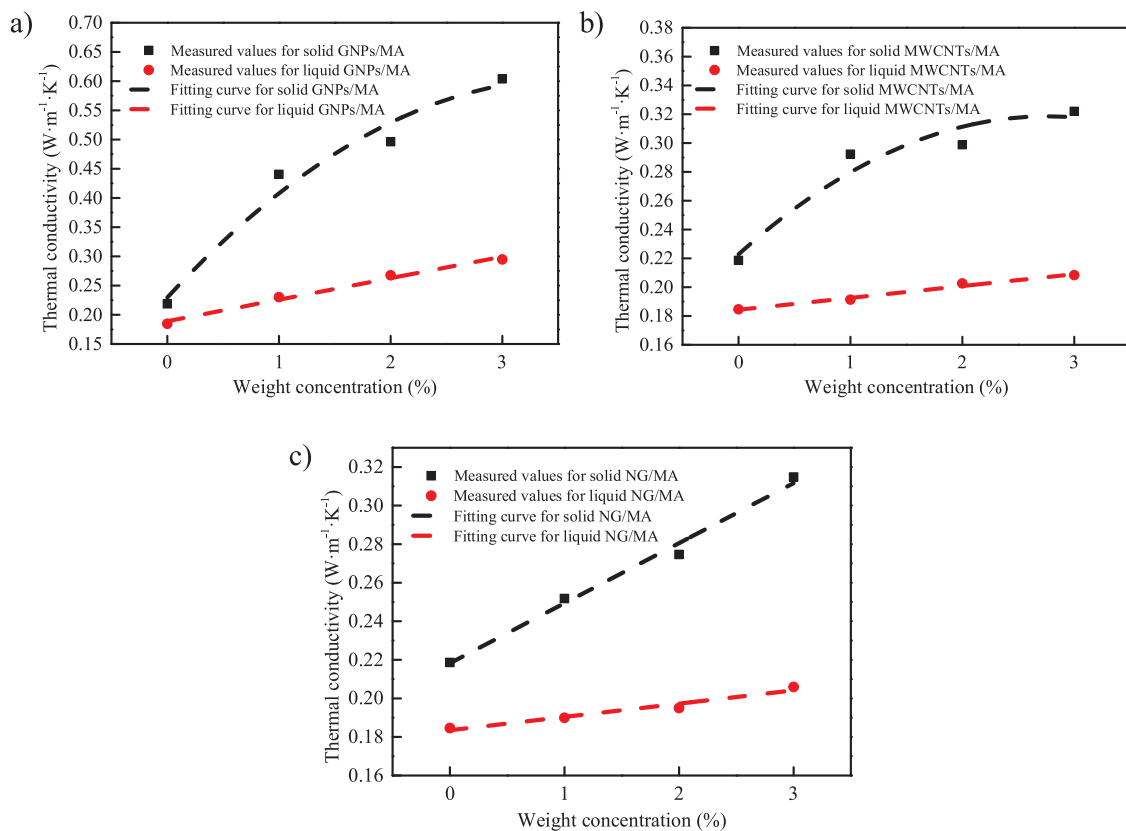


Fig. 6. Effect of the weight concentration of nano-additives on thermal conductivity for: a) GNPs/MA PCM composites; b) MWCNTs/MA PCM composites; and c) NG/MA PCM composites.

Table 3
The thermal conductivity of the samples.

Samples	Thermal conductivity [W·m ⁻¹ ·K ⁻¹]		Relative improvements [%]	
	Solid	Liquid	Solid	Liquid
MA	0.2186	0.1846	–	–
1 wt% GNPs/MA	0.4405	0.2302	101.51	24.70
2 wt% GNPs/MA	0.4963	0.2677	127.04	45.02
3 wt% GNPs/MA	0.6039	0.2948	176.26	59.70
1 wt% MWCNTs/MA	0.2923	0.1913	33.71	3.63
2 wt% MWCNTs/MA	0.2988	0.2026	36.69	9.75
3 wt% MWCNTs/MA	0.3220	0.2083	47.30	12.84
1 wt% NG/MA	0.2518	0.1899	15.19	2.87
2 wt% NG/MA	0.2746	0.1950	25.62	5.63
3 wt% NG/MA	0.3148	0.2059	44.01	11.54

between nanoparticles within the percolation network, due to the high thermal resistivity associated with low-frequency phonons transfer at the interface [22,41–44]. The special flake-like configuration for the GNPs enabled the fewer interfaces in the percolation network, compared to the wire-like configuration for the MWCNTs, which therefore better facilitated the thermal conductivity enhancement.

Different from the GNPs/MA and MWCNTs/MA composites, the linear increasing trends in the thermal conductivity were found for the NG/MA under both solid and liquid phases. When adding 3 wt% NG into MA, the thermal conductivity of the NG/MA composite was improved linearly to 0.3148 W·m⁻¹·K⁻¹ and 0.2059 W·m⁻¹·K⁻¹ under the solid and liquid phases, respectively. It was only enhanced by 44.01% under the solid phase, which was much lower than that of the GNPs/MA composite. This might be due to the existence of a threshold for the nanoparticles, above which effective mutual interlinkage among the nanoparticles can be achieved to establish the percolating network [45,46]. This threshold was highly dependent on the aspect ratio of the nanoparticles. Since the aspect ratio of the NG particles was much smaller, a higher weight concentration was required to establish the percolation network and achieve the same level thermal conductivity enhancement effect as that of the GNPs.

To gain an insight of the temperature-dependency of the thermal conductivity enhancement, the thermal conductivity of MA, 3 wt% GNPs/MA, 3 wt% MWCNTs/MA and 3 wt% NG/MA was further measured under different temperatures, as presented in Fig. 7. Similar variation trends in the thermal conductivity with the variation of the temperature were found for the three PCM composites. The thermal conductivity was relatively constant in solid phase below 50 °C for the individual PCM composites. However, a step decrease in the thermal conductivity was observed when increasing the temperature across the

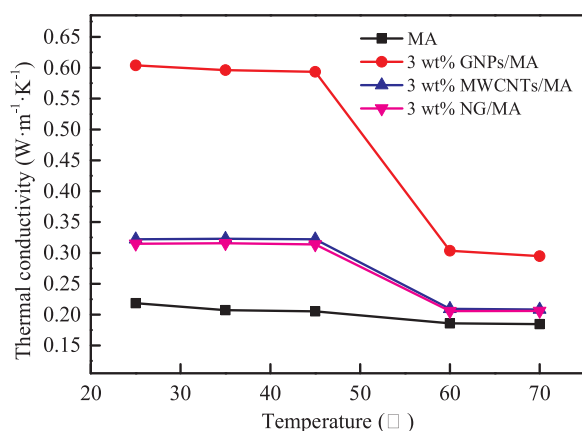


Fig. 7. Thermal conductivity of MA, 3 wt% GNPs/MA, 3 wt% MWCNTs/MA, and 3 wt% NG/MA as a function of temperature.

phase transition temperature range. When the temperature was above 60 °C in the liquid phase, the thermal conductivity became relatively constant again, but it was substantially lower than that under the solid phase for each PCM composite. This phenomenon was due to the random molecular motion in the liquid phase, which collapsed the percolation network established by nanoparticles under the solid phase. This result was in agreement with the work of Qian et al. [47] and Babaei et al. [48].

Base on the above discussion, it can be concluded that the carbon-based nanoparticles with inherent high thermal conductivity (reported to be as high as 3000–6600 W·m⁻¹·K⁻¹) [23–25] can effectively improve the thermal conductivity of PCM composites. However, to maximise the enhancement in the thermal conductivity of the PCM composites, the concentration rate, configuration of the nanoparticles, and dispersion of the nano-additives need to be carefully selected and handled, and the attention needs to be paid on the solid/liquid state of the PCM.

3.4. DSC analysis

The DSC curves of the raw MA and GNPs/MA PCM composites with different additive weight concentrations are presented in Fig. 8a, and the corresponding phase change temperature and latent heat are summarized in Table 4. It can be seen that the GNPs/MA composites experienced sharp phase change processes during both melting and freezing processes (Fig. 8a), compared with that of the raw MA which had clearly wider phase change temperature ranges. The on-set temperatures of the melting process for GNPs/MA PCM composites with the GNPs concentration of 1 wt%, 2 wt%, and 3 wt% were the same of 54.3 °C, while that of the freezing process were 54.2 °C, 54.4 °C and 54.4 °C, respectively. As a comparison, the raw MA had the on-set temperatures of 54.4 °C for the melting process, and 52.4 °C for the freezing process. It demonstrated the effectiveness of using GNPs to modify the phase change characteristics of MA, which approached a preferable isothermal phase change process. It can also be concluded that this effect was not sensitive to the concentration of the GNPs added. In the meanwhile, only a slight decrease in the latent heat was observed for the GNPs/MA composites when compared with the raw MA. The latent heat for the 1 wt%, 2 wt% and 3 wt% GNPs/MA composites were 189.18 J·g⁻¹, 187.39 J·g⁻¹ and 187.19 J·g⁻¹ for melting, and 190.88 J·g⁻¹, 189.70 J·g⁻¹ and 188.58 J·g⁻¹ for freezing, respectively, which were only reduced by 2.93%, 3.85% and 3.96% (for melting) and 2.19%, 2.79% and 3.37% (for freezing). It can also be seen that by adding the GNPs, the supercooling during the freezing process of MA was effectively eliminated (Fig. 8a), due to the nucleating effect of the nano-additives [49] which served as nucleating agents to facilitate the crystallization of MA.

Similar to the GNPs/MA PCM composites, narrow endothermic peaks and exothermic peaks were observed in the DSC curves for the MWCNTs/MA composites, compared with the raw MA (Fig. 8b). The adding of MWCNTs also showed a minor influence on the latent heat of the PCM composites. By adding 1 wt%, 2 wt% and 3 wt% MWCNTs into MA, the latent heat of fusion decreased from 194.90 J·g⁻¹ to 191.99 J·g⁻¹, 190.12 J·g⁻¹ and 188.47 J·g⁻¹, while the latent heat of solidification dropped from 195.15 J·g⁻¹ to 192.19 J·g⁻¹, 190.99 J·g⁻¹ and 188.21 J·g⁻¹, respectively. The phenomenon with smaller phase change temperature ranges can also be found when adding NG into MA, together with the minor influence on the latent heat of the NG/MA composites, as presented in Fig. 8c and Table 4. It seemed that the three carbon-based nano-additives had a similar effect in terms of modifying the phase change process of MA.

3.5. TGA analysis

Fig. 9 presents the TGA profiles for PCM composites modified by different carbon-based nano-additives, and the details are further

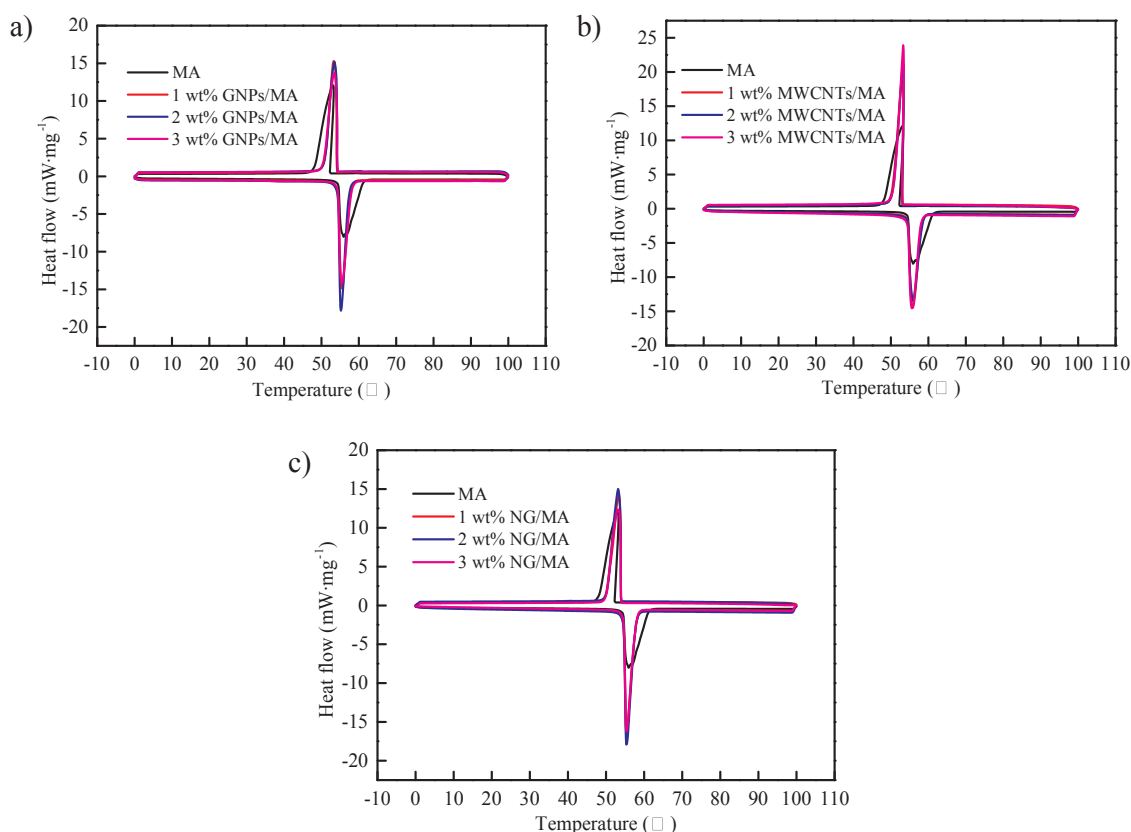


Fig. 8. DSC melting and freezing curves for: a) GNPs/MA PCM composites; b) MWCNTs/MA PCM composites; and c) NG/MA PCM composites.

Table 4

Phase change temperature and phase change latent heat of GNPs/MA, MWCNTs/MA and NG/MA PCM composites.

Samples	On-set temperature (°C)		Latent heat ($\text{J}\cdot\text{g}^{-1}$)	
	Melting	Freezing	Melting	Freezing
MA	54.4	52.4	194.90	195.15
1 wt% GNPs/MA	54.3	54.2	189.18	190.88
2 wt% GNPs/MA	54.3	54.4	187.39	189.70
3 wt% GNPs/MA	54.3	54.4	187.19	188.58
1 wt% MWCNTs/MA	54.6	53.2	191.99	192.19
2 wt% MWCNTs/MA	54.4	53.1	190.12	190.99
3 wt% MWCNTs/MA	54.4	53.0	188.47	188.21
1 wt% NG/MA	54.5	54.1	191.27	191.15
2 wt% NG/MA	54.6	54.0	190.67	191.04
3 wt% NG/MA	54.6	54.1	188.90	188.85

summarized in Table 5. By adding the three types of nano-additives into MA, lower initial decomposition temperatures were found, ranging from 116.0 °C to 142.8 °C with different carbon-based nanoparticles concentrations, compared with that of 152.3 °C for the raw MA. The decrease in the initial decomposition temperature was acceptable, as it was far higher than the phase change temperature range of MA, even though it reduced the safe operation temperature limits for the application of the NePCM composites. Similar to the initial decomposition temperature, the lower maximal and final decomposition temperatures were observed for the PCM composites modified with the carbon-based nanoparticles additives.

3.6. Exothermic process analysis

Fig. 10 presents the thermal exothermic process of the bulk MA, 3 wt% GNPs/MA, 3 wt% MWCNTs/MA and 3 wt% NG/MA PCM

composites. It can be seen that with the modification of 3 wt% GNPs, the discharging time of GNPs/MA composite was significantly reduced to around 4900 s, compared to that of around 7450 s for the raw MA. Similarly, the adding of MWCNTs with a concentration of 3 wt% substantially decreased the discharging time of MWCNTs/MA composite to around 5850 s. When using NG particles as additives with a concentration of 3 wt%, the discharging time of the PCM composite can also be reduced to around 6500 s. The performance of the bulk PCM composites reflected the thermal conductivity enhancement effect due to the adding of different carbon-based nano-additives. The GNPs/MA composites whose thermal conductivity enhancement outperformed the other two PCM composites provided the best bulk PCM discharging performance with the shortest discharging completion time.

3.7. Thermal cycling stability analysis

Fig. 11 compares the DSC curves of different PCM composite samples after per 100 thermal cycles, and Table 6 further summarizes the corresponding phase change characteristics before and after the thermal cycles. It can be seen from Fig. 11a that the DSC curves of the 3 wt% GNPs/MA PCM composite after 100, 200 and 300 thermal cycles were consistent with the original DSC curve before thermal cycling, and no obvious degradation can be observed. The corresponding latent heat of fusion was 185.69 $\text{J}\cdot\text{g}^{-1}$ after 100 thermal cycles, 183.43 $\text{J}\cdot\text{g}^{-1}$ after 200 thermal cycles, and 181.75 $\text{J}\cdot\text{g}^{-1}$ after 300 thermal cycles, respectively, which were only reduced by 0.80%, 2.01%, and 2.91% in comparison with the sample before the thermal cycling test. Slight decrease was also observed for the latent heat of solidification with the increasing number of thermal cycles, but it was still above 180 $\text{J}\cdot\text{g}^{-1}$ after 300 thermal cycles (i.e. 180.44 $\text{J}\cdot\text{g}^{-1}$) with a degradation no more than 5% (i.e. 4.32%). Similarly, negligible degradation in the latent heat was found in 3 wt% MWCNTs/MA composite sample (see Fig. 11b) and 3 wt% NG/MA composite sample (see Fig. 11c) after the thermal

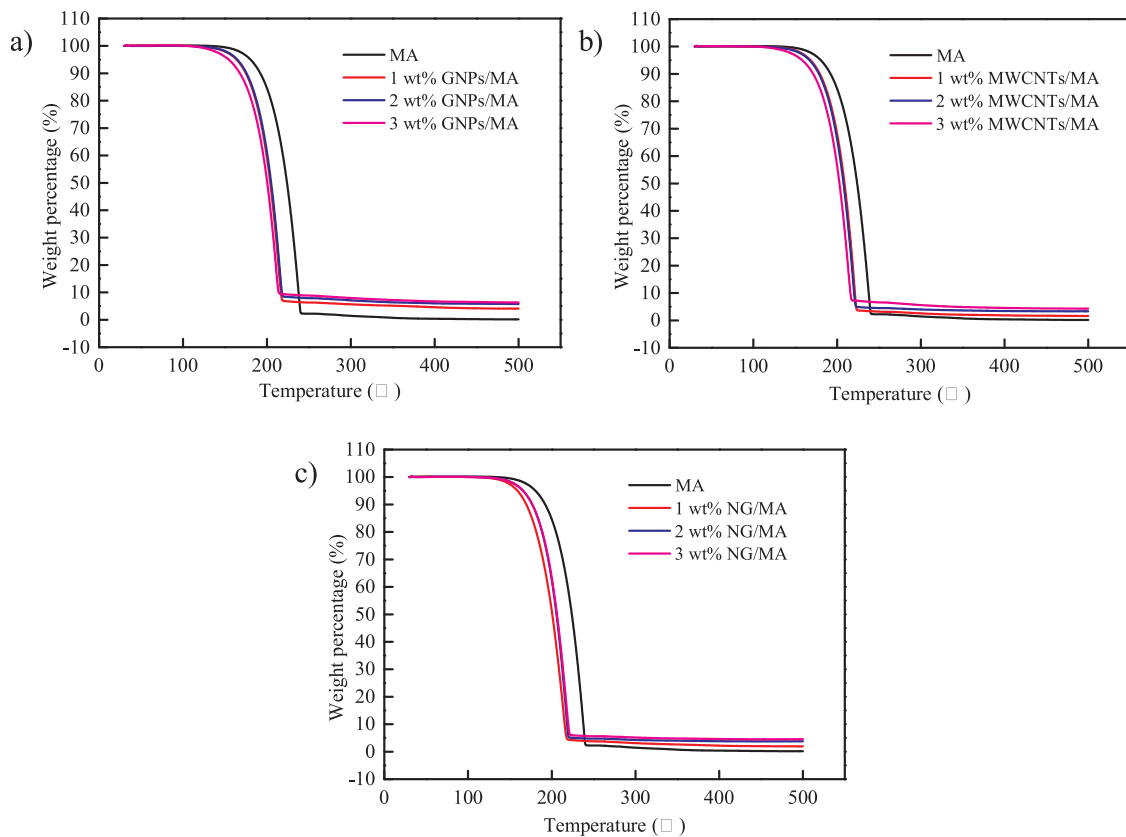


Fig. 9. TGA profiles for: a) MA and GNPs/MA PCM composites; b) MA and MWCNTs/MA PCM composites; and c) MA and NG/MA PCM composites.

Table 5

TGA data for MA, GNPs/MA, MWCNTs/MA and NG/MA PCM composites.

Samples	Initial decomposition temperature (°C)	Maximal decomposition temperature (°C)	Final decomposition temperature (°C)
MA	152.3	238.2	241.5
1 wt% GNPs/MA	130.2	214.5	219.5
2 wt% GNPs/MA	130.2	215.5	220.2
3 wt% GNPs/MA	116.0	210.8	215.7
1 wt% MWCNTs/MA	142.8	221.0	224.5
2 wt% MWCNTs/MA	133.3	219.0	223.3
3 wt% MWCNTs/MA	118.0	213.5	218.3
1 wt% NG/MA	130.0	214.5	220.2
2 wt% NG/MA	132.2	216.8	221.0
3 wt% NG/MA	132.3	217.7	223.3

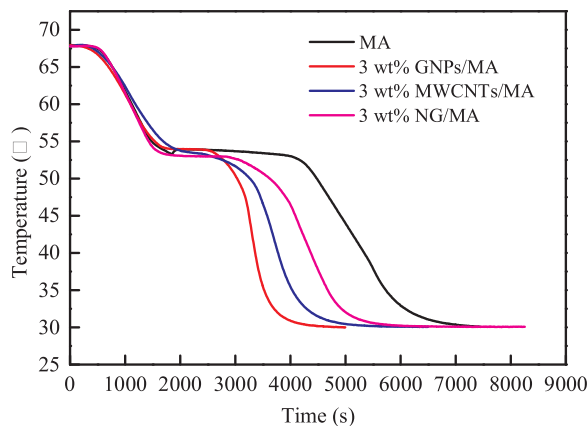


Fig. 10. Cooling curves of MA, 3 wt% GNPs/MA, 3 wt% MWCNTs/MA and 3 wt% NG/MA PCM composites.

cycling tests, even though a slight decreasing trend was found when the number of thermal cycles increased. The above results indicated that the nano-enhanced PCM composites prepared had excellent thermal cycling stabilities.

To further obtain a clear insight into the thermal cycling stability, the crystallization characteristics of the PCM composite samples were studied before and after 300 thermal cycling tests, as presented in Fig. 12. It can be seen that the two XRD curves were consistent with each other for the GNPs/MA composite (Fig. 12a), with identical absorption peaks at the diffraction angles (2θ) of 5.6° , 8.3° , 14.0° , 21.5° and 24.1° . It indicated that after 300 thermal cycles, the crystalline structure of 3 wt% GNPs/MA composite sample remained the same. Similar results can be concluded by comparing the XRD curves of 3 wt% MWCNTs/MA (see Fig. 12b) and 3 wt% NG/MA (see Fig. 12c) before and after the thermal cycling test. By further analysing the FT-IR spectra before and after 300 thermal cycles of the PCM composites (Fig. 13), it was found that the chemical structures of the PCM composites also remained stable.

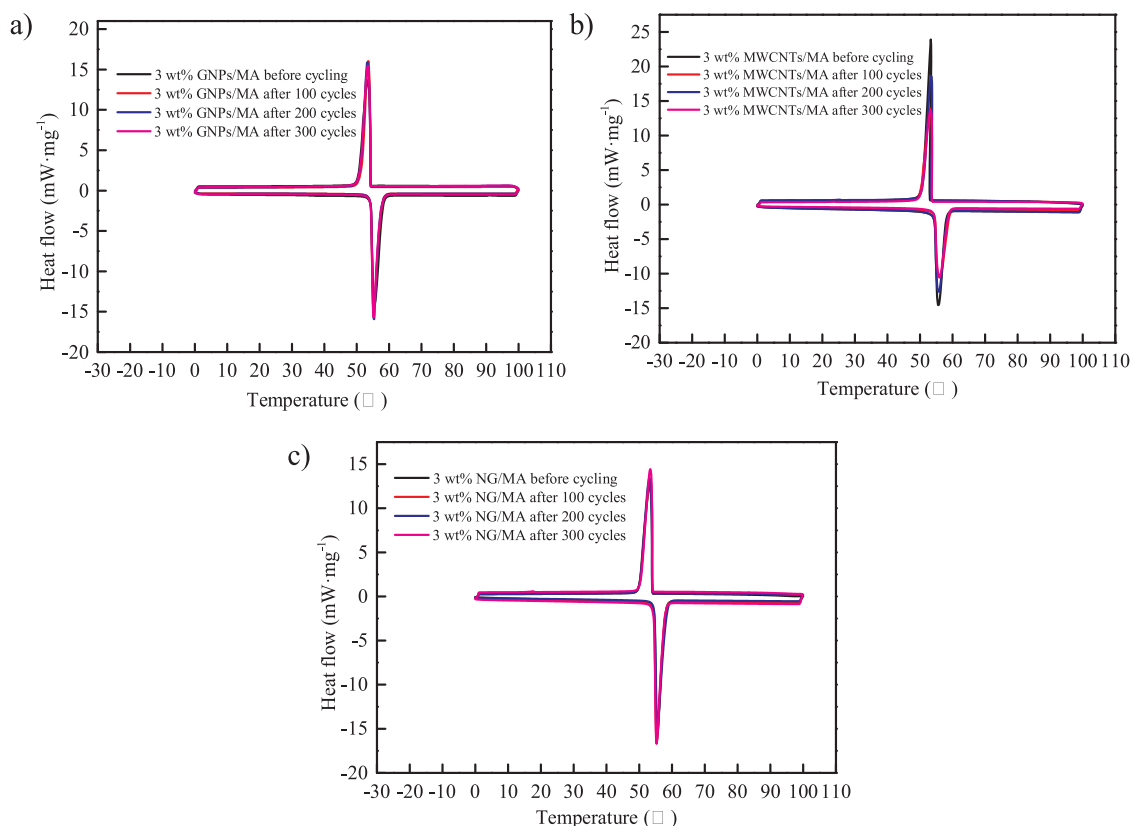


Fig. 11. DSC melting and freezing curves after 0, 100, 200 and 300 thermal cycles for: a) 3 wt% GNPs/MA PCM composite; b) 3 wt% MWCNTs/MA PCM composite; and c) 3 wt% NG/MA PCM composite.

Table 6

Phase change characteristics of thermal cycles of 3 wt% GNPs/MA, 3 wt% MWCNTs/MA and 3 wt% NG/MA PCM composites.

Samples	Number of thermal cycles	On-set temperature (°C)		Latent heat (J·g ⁻¹)	
		Melting	Freezing	Melting	Freezing
3 wt% GNPs/MA	0	54.3	54.4	187.19	188.58
	100	54.3	54.3	185.69	183.00
	200	54.5	54.1	183.43	181.83
	300	54.4	54.2	181.75	180.44
3 wt% MWCNTs/MA	0	54.4	53.0	188.47	188.21
	100	54.5	53.6	185.77	186.32
	200	54.3	53.5	184.85	184.97
	300	54.3	53.8	181.21	182.86
3 wt% NG/MA	0	54.6	54.1	188.90	188.85
	100	54.5	54.1	186.02	187.13
	200	54.6	54.0	184.38	185.93
	300	54.4	54.1	182.68	182.03

4. Conclusions

This paper presented the preparation, thermal characterization and thermal examination of phase change materials (PCMs) enhanced by carbon-based nano-particles for solar thermal energy storage applications, by following a proposed experimental framework for the development of nano-enhanced PCMs (NePCMs). The framework enabled the comprehensive assessment of the prepared NePCMs before practical applications. Myristic acid (MA) was selected as the base PCM, and three different types of nanoparticles, *i.e.* graphene nanoplatelets (GNPs), multi-walled carbon nanotubes (MWCNTs) and nano-graphite (NG), were used as nano-additives to modify the properties of the base PCM. The properties of the nanoparticles/PCM composites, including morphology, chemical stability, and thermal properties were measured

to characterize the NePCMs. The thermal cycling stability and bulk thermal performance of the NePCMs prepared were also tested.

It was found that the thermal conductivity of the MA can be significantly enhanced by adding carbon-based nanoparticles as additives. Using GNPs as nano-additives showed the best performance for thermal conductivity enhancement, which can significantly enhance the thermal conductivity by 176.26% under the solid phase, compared to that of 47.30% and 44.01% for MWCNTs and NG, under the weight concentration of 3 wt%. The high thermal conductivity in the GNPs/MA composite was due to the thermal conductive percolation network established by GNPs, which tended to have fewer nanoparticle interfaces, compared to the other carbon-based nanoparticles. Different increasing trends of the thermal conductivity variation under solid and liquid phases were found for both GNPs/MA and MWCNTs/MA PCM composites, which were nonlinear and linear, respectively. However, due to the existence of random molecular motion under the liquid phase, the corresponding thermal conductivity enhancement was evidently lower than that under the solid phase. To establish and maintain an effective percolation network, the concentration of nano-additives needs to be carefully selected. It was also found that by adding the carbon-based nanoparticles as additives into the MA, the phase transition can be modified to approach an isothermal process and the supercooling effect can be eliminated, while still maintaining a high latent heat capacity. The thermal cycling and bulk thermal performance tests further demonstrated the practical performance of the NePCMs developed, which exhibited excellent thermal cycling stability, and great potential to reduce the phase change time required. The NePCMs developed can be used to develop solar thermal energy storage for various applications.

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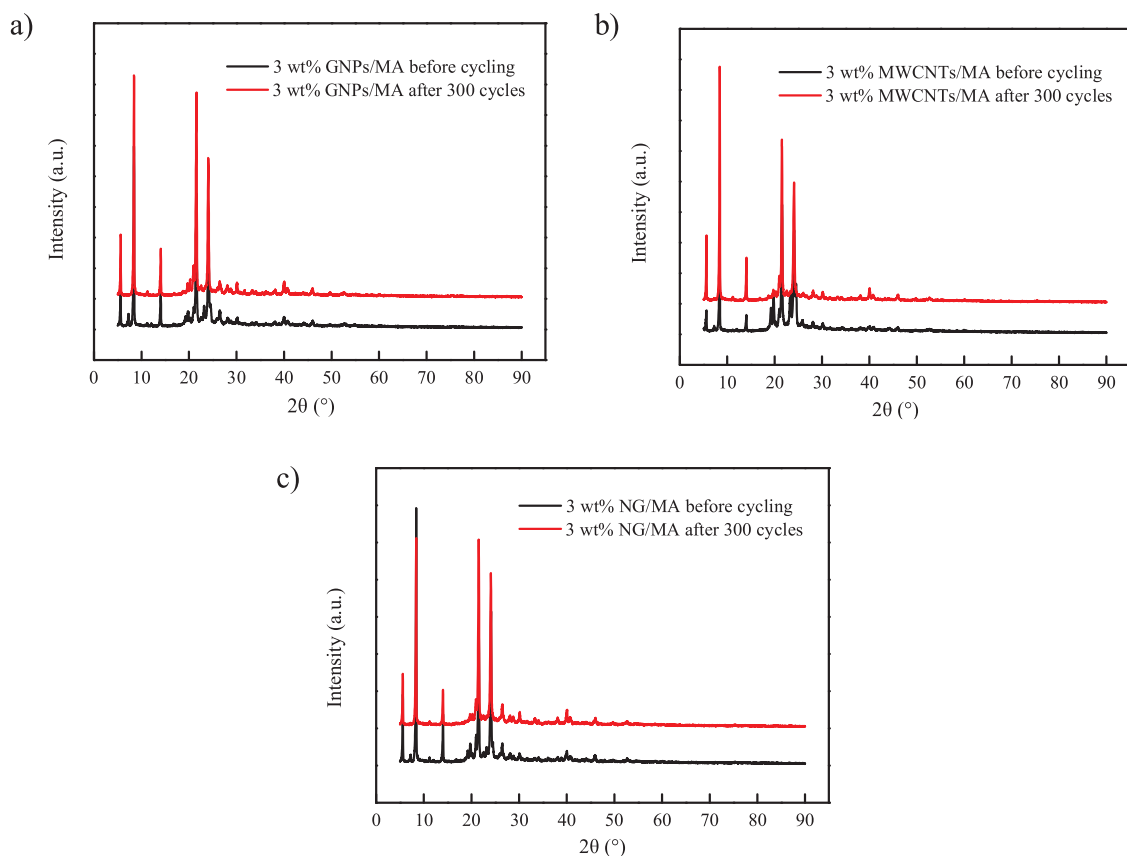


Fig. 12. XRD spectra before and after cycling for: a) 3 wt% GNPs/MA PCM composite; b) 3 wt% MWCNTs/MA PCM composite; and c) 3 wt% NG/MA PCM composite.

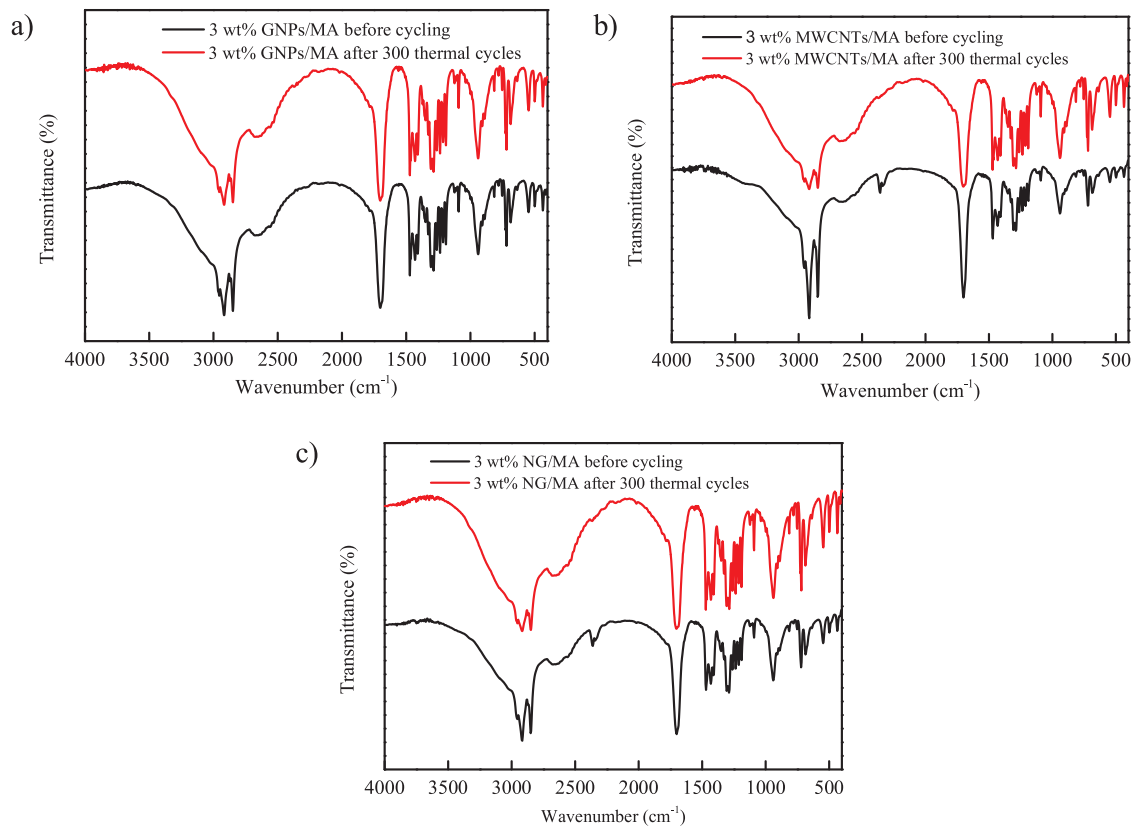


Fig. 13. FT-IR spectra before and after cycling for: a) 3 wt% GNPs/MA PCM composite; b) 3 wt% MWCNTs/MA PCM composite; and c) 3 wt% NG/MA PCM composite.

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